

AirSense

Final R&D; Problem Statement

Problem Statement

When a citizen reports a pollution incident anywhere in India, no system today closes the loop between that complaint and a measured result. CPCB's own SAMEER app proves citizens will report incidents at volume — but every report is then routed manually, with no automated diagnosis of what caused it, no estimate of who's affected, and no record of whether the response actually worked.

AirSense will build that missing intelligence layer for Indian cities: for a reported pollution incident, automatically infer the probable source category, estimate the wind-adjusted affected population zone, recommend an evidence-based intervention, and measure whether AQI actually improved afterward. Starting with construction and road dust — the source category CPCB's and TERI's own studies flag as the dominant contributor to PM10 across Indian cities (20–66% depending on location and season) — the system will generalize to other source types as the pipeline matures.

The target users are dual: the citizen who filed the complaint and deserves transparency, and the municipal nodal officer who is legally required to resolve SAMEER complaints within 24 hours but currently has no decision-support tool to do so.

Key Differentiators

- Closes the loop between complaint and measured outcome — the rarest capability in India's air quality landscape
- Evidence-based, not assumptive: SAMEER app exists nationally with documented complaint volume and unresolved complaints
- Dual-sided value: serves both citizens (transparency) and municipal officers (decision-support for 24-hour resolution)
- Narrow wedge, broad scale: starts with construction/road dust (nationally dominant), generalizes to other sources
- Defensible positioning: doesn't compete with SAFAR, CAMS, PRANA on forecasting — owns the complaint-to-outcome layer

What Changed from the Original Statement

This version maintains the core insight while making it nationally scalable. Here's what evolved:

Change	Rationale
'No system closes the loop' replaced generic finding	Now evidenced by SAMEER's 2.4★ rating and unresolved-complaint reviews — documented nationally
Construction/road dust as first vertical (nationally)	CPCB/TERI studies flag this as dominant PM10 source (20–66%) across Indian cities
'Measure whether it worked' as co-equal objective	Rarest capability in the landscape — not even PRANA automates this — now positioned as key differentiator
Nodal officer as second target user	Person legally required to resolve SAMEER complaints in 24 hours — stronger first citizen focus
Deliberately excluded forecasting, satellite, etc.	SAFAR, CAMS, Blue Sky Analytics already win there — scope creep would drag into other domains

Why This Works for National Implementation

Evidence-Based, Not Assumptive

SAMEER app exists nationally (Android + iOS) with documented complaint volume. CPCB guidelines mandate 24-hour resolution. You're solving a documented operational gap in an existing national system.

Narrow Enough to Build, Broad Enough to Scale

Starting vertical: construction/road dust (visually obvious, high complaint volume, validate against PM10 + wind + nearby sensor). Generalization path: extend to biomass burning, industrial emissions, vehicular pollution using same inference logic. Road dust contributes 20–66% of PM10 across Indian cities (Delhi NCR, Kolkata, Lucknow, etc.).

Defensible Differentiation

SAFAR, CAMS, PRANA already win on measurement and forecasting — you're not competing there. Your unique value: closing the loop between complaint → diagnosis → intervention → measured outcome. Nobody automates outcome measurement — not even PRANA at national level.

Dual-Sided Value Proposition

Citizen: transparency, accountability, proof that their complaint mattered. Nodal Officer: decision-support tool to meet legally mandated 24-hour resolution deadline. This makes you valuable to both the regulated and the regulator — stronger position than citizen-only tool.

One-Sentence Elevator Pitch

AirSense closes the loop on India's air pollution complaints: for every SAMEER app report, we automatically diagnose the source, estimate who's affected, recommend an intervention, and measure whether it actually worked — starting with construction and road dust, the dominant pollution source across Indian cities.

Implementation Note for Proposals

While the initial pilot will focus on construction and road dust incidents in your city/region, the system architecture is designed for national deployment. The inference engine is source-category-agnostic (dust → biomass → industrial → vehicular), the meteorological integration uses IMD's national API coverage, and the intervention recommendation library can be extended city-by-city as municipal partnerships scale. This mirrors the national rollout strategy of SAMEER itself, which started in Delhi NCR and expanded to 100+ cities.

Sources

CPCB SAMEER App (Android/iOS), CPCB Guidelines for Air Pollution Complaints, TERI Source Apportionment Studies (Delhi NCR, Kolkata, Lucknow), CEEW Air Quality Reports, National Clean Air Programme (NCAP) Progress Reports